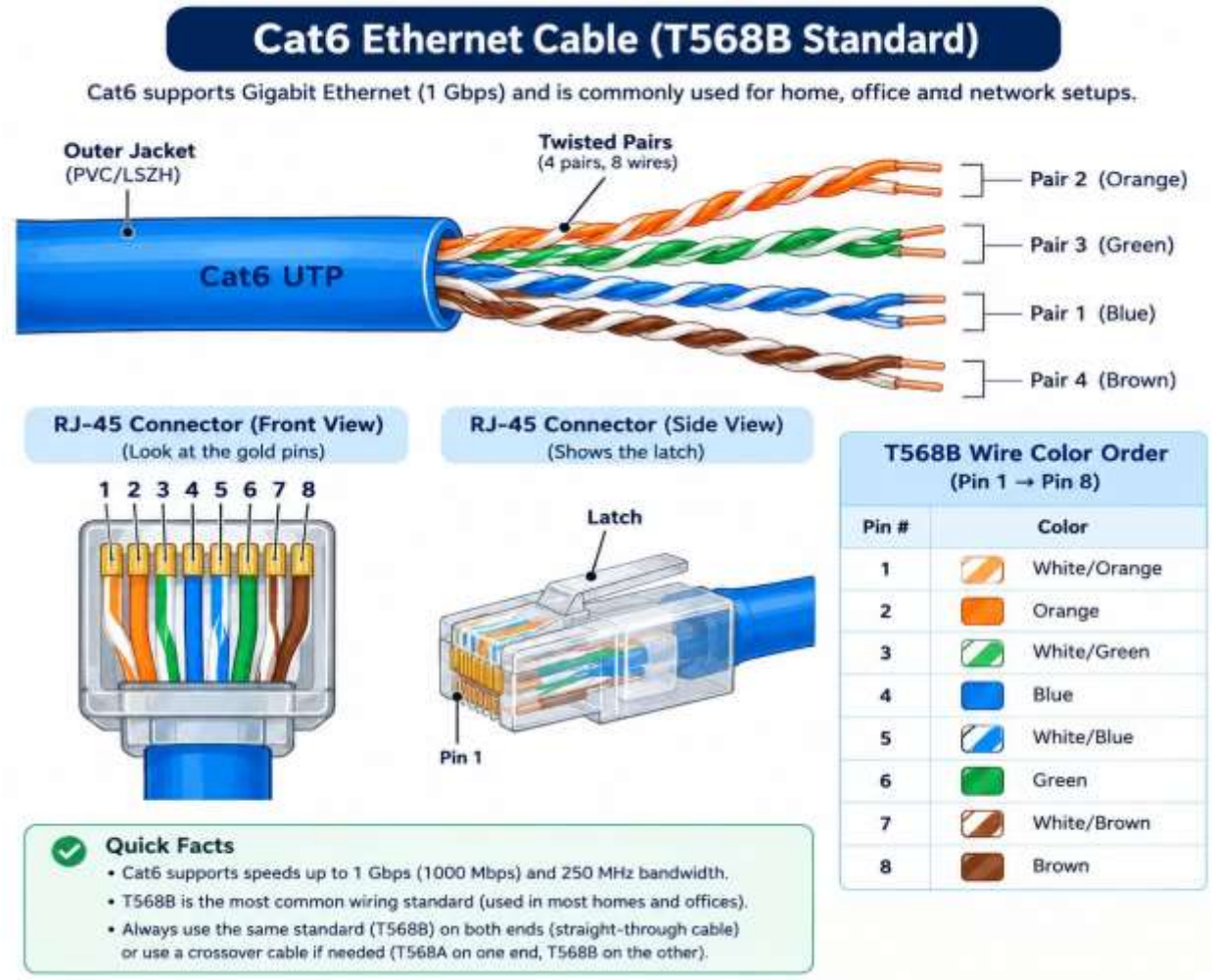
NETWORK CABLE AND CONNETORS

TASK-1 Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



NETWORK CABLE AND CONNETORS

TASK-2 List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

No.	Cable Type	Real-World Device / Application	Why It Is Used
1	Cat5e Ethernet Cable	Wi-Fi router connected to a computer	Provides a reliable wired connection for internet access at home or in a small office.
2	Cat6 Ethernet Cable	Zomato restaurant POS terminal	Supports high-speed wired communication for processing orders and updating billing information.
3	Fiber Optic Cable	IPL streaming server	Supports high bandwidth and carries data over long distances, helping deliver live video streams efficiently.

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TASK-3 Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Step-by-step process

1. Cut the cable: Cut the Ethernet cable to the required length using a cable cutter.
2. Strip the outer jacket: Remove about 2–3 cm of the outer covering carefully without damaging the inner wires.
3. Arrange the wires: Straighten the eight wires and arrange them according to the T568B standard.
4. Follow the colour order: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
5. Trim the wires: Cut all eight wires evenly so they fit properly inside the RJ-45 connector.
6. Insert the wires: Push the wires fully into the connector, ensuring each wire reaches the end and remains in the correct position.
7. Crimp the connector: Place the connector into the crimping tool and press firmly to secure the wires and contacts.
8. Test the cable: Use a LAN cable tester to check that all eight pins are connected correctly.

Two common mistakes to avoid

1. Incorrect wire colour order: Mixing up the T568B sequence can cause the cable to fail or work incorrectly.
2. Wires not fully inserted: If the wires do not reach the connector contacts, the connection may fail or become intermittent.

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TASK-4 Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Connection	Recommended cable	Connector type	Reason
Gaming console to TV/monitor	High-Speed HDMI 2.0 or Ultra High-Speed HDMI 2.1	HDMI Type-A	Carries high-quality video and audio with low latency.
Router or network switch to consoles/PCs	Cat6 Ethernet cable	RJ-45	Provides a stable, low-latency network connection for online gaming.
Main router to network switch	Cat6 Ethernet; fiber for long-distance links if needed	RJ-45 or fiber-optic connector	Connects multiple devices reliably and supports high-speed data transfer.
Projector or large display	HDMI cable; active HDMI or fiber HDMI for long runs	HDMI Type-A	Delivers clear video and audio to the main screening display.
Console or display to electricity supply	Manufacturer-recommended power cable	Compatible power plug	Supplies safe, reliable power to each device.

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TASK-5 Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a Wi Fi router or at a café's POS counter), and describe what type it is and what devices it connects. Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



Description of the photo

- Cable type: Ethernet cable (Cat5e or Cat6; the exact category cannot be confirmed from the photo).
- Connector type: RJ-45 connector.
- Devices connected: A Wi-Fi router and a computer or another network device at the other end.
- Purpose: It transfers data between devices and provides a stable, high-speed wired network connection.